

JEE MAIN REVISION MASTER: ROTATIONAL MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Torque, Angular Momentum & Pure Rolling	8 PYQs	Dynamics & Rolling
Moderate	Moment of Inertia & Parallel/Perpendicular Axes Theorems	6 PYQs	Inertia

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Torque, Angular Momentum & Pure Rolling	$\vec{\tau} = I\vec{\alpha} = \frac{d\vec{L}}{dt}$, $K_{\text{roll}} = \frac{1}{2}Mv_{\text{cm}}^2 \left(1 + \frac{k^2}{R^2}\right)$, $a = \frac{g \sin \theta}{1 + k^2/R^2}$
Moment of Inertia & Parallel/Perpendicular Axes Theorems	$I = \int r^2 dm$, $I = I_{\text{cm}} + Md^2$, $I_z = I_x + I_y$ (laminar planar bodies)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Torque, Angular Momentum & Pure Rolling (High Yield — 8 PYQs)

Core Principle: Rotational analogue of Newton's second law, conservation of angular momentum, and rolling down an incline.

■ **Caution / JEE Pitfall:** On an incline, the body with the smallest k^2/R^2 reaches the bottom first (Solid Sphere > Solid Cylinder > Hollow Sphere > Ring).

■ **Speed Trick:** In pure rolling on a flat surface, the instantaneous axis of rotation passes through the contact point ($v_{\text{contact}} = 0$).

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The angular momentum of a moving body remains constant, if..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1 = 4 \text{ kgm}^2, r_1 = 20 \text{ cm}$, $I_2 = 20 \text{ kgm}^2$ and $r_2 = 30 \text{ cm}$..." [Ans: N/A]

3.2 Moment of Inertia & Parallel/Perpendicular Axes Theorems (Moderate Yield — 6 PYQs)

Core Principle: Calculation of rotational inertia, radius of gyration $k = \sqrt{I/M}$, and axes shift theorems.

■ **Caution / JEE Pitfall:** Perpendicular axis theorem $I_z = I_x + I_y$ is strictly valid ONLY for 2D laminar (flat) objects lying in the xy-plane.

■ **Speed Trick:** Solid sphere $I_{cm} = (2/5)MR^2$; hollow spherical shell $I_{cm} = (2/3)MR^2$; circular disc $I_{cm} = (1/2)MR^2$.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1 = 4 \text{ kgm}^2, r_1 = 20 \text{ cm}$, $I_2 = 20 \text{ kgm}^2$ and $r_2 = 30 \text{ cm}$..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The moment of inertia of a rigid body about an axis..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Seven identical discs are arranged in a planar pattern, so as to touch each other as shown in the figure. Each disc has mass m radius R ..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If the earth were to suddenly contract to half of its present radius, what would be the duration of the day?

- (A) 6 h
- (B) 18 h
- (C) 24 h
- (D) 30 h

Correct Answer: None

Solution Analysis: When the earth contracts suddenly, no external torque is acting on it. So its angular momentum remains conserved. Angular momentum is $L = I\omega$. For a spherical body, $I \propto R^2$. If the radius becomes half, $R' = (R/2)$. So the new moment of inertia becomes $I' = I((1/2))^2 = I/4$. Using conservation of angular momentum, $I\omega = I'\omega'$. $I\omega = (I/4)\omega'$. $\omega' = 4\omega$. So the earth will rotate 4 times faster. Time period is inversely proportional to angular velocity, so $T' = (T/4)$. Present duration of a day is $T = 24 \text{ h}$. Hence, $T' = 6 \text{ h}$.

Benchmark Problem 2: KCET 2025 (Q#3)

Two fly wheels are connected by a non-slipping belt as shown in the figure. $I_1 = 4 \text{ kgm}^2, r_1 = 20 \text{ cm}$, $I_2 = 20 \text{ kgm}^2$ and $r_2 = 30 \text{ cm}$. A torque of 10 Nm is applied on the smaller wheel. Then match the entries of column I with appropriate entries of column II. I Quantities II Their numerical Values (in SI units) (a) Angular acceleration of smaller wheel (i) $(5/3)$ (b) Torque on the larger wheel (ii) $(100/3)$ (c) Angular acceleration of larger wheel (iii) $(5/2)$

- (A) a-ii, b-iii, c-i
- (B) a-iii, b-i, c-ii
- (C) a-ii, b-i, c-iii
- (D) a-iii, b-ii, c-i

Correct Answer: None

Solution Analysis: Explanation The relationship between the torque, moment of inertia, and angular acceleration for each wheel can be established with the following equations: For the smaller wheel: $T_1 = I_1 \alpha_1$ where T_1 is the torque applied to the smaller wheel, $I_1 = 4 \text{ kg} \cdot \text{m}^2$ is its moment of inertia, and α_1 is the angular acceleration. The angular acceleration ratio between the two wheels is given by: $\alpha_2 r_2 = \alpha_1 r_1$ where α_2 is the angular acceleration of the larger wheel, $r_1 = 0.2 \text{ m}$, and $r_2 = 0.3 \text{ m}$. Given that